

# PCA (principal component Analysis)

curse of dimensionality  
visualization

> PCA

multiple feature  $\rightarrow$  PC convert

| X | Y |
|---|---|
| 2 | 1 |
| 3 | 4 |
| 5 | 2 |
| 6 | 5 |

100%

① standardization - mean centering

$$\bar{X} = \frac{2+3+5+6}{4} = 4.0$$

$$\bar{Y} = \frac{1+4+2+5}{4} = 3.0$$

| X | $\bar{X}$ | Y | $\bar{Y}$ | $X_i - \bar{X}$ | $Y_i - \bar{Y}$ |
|---|-----------|---|-----------|-----------------|-----------------|
| 2 | 4         | 1 | 3         | $2 - 4 = -2$    | $1 - 3 = -2$    |
| 3 |           | 4 |           | $3 - 4 = -1$    | $4 - 3 = 1$     |
| 5 |           | 2 |           | $5 - 4 = 1$     | $2 - 3 = -1$    |
| 6 |           | 5 |           | $6 - 4 = 2$     | $5 - 3 = 2$     |

## ② Covariance matrix

$$\text{cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$\begin{aligned} \cdot \text{Cov}(X, X) &= \sum (x_i - \bar{x})^2 / n-1 = \text{variance of } x \\ \text{Cov}(Y, Y) &= \sum (y_i - \bar{y})^2 / n-1 = \text{variance of } y \\ \cdot \text{Cov}(X, Y) &= \sum (x_i - \bar{x})(y_i - \bar{y}) / n-1 = \text{covariance} \end{aligned}$$

$$\begin{array}{cc} \text{Cov}(X, X) = & \\ \begin{array}{cc} x & x^2 \\ -2 & 4 \\ -1 & 1 \\ 1 & 1 \\ 2 & 4 \end{array} & \frac{10}{4-1} = \boxed{3.33} \end{array}$$

$$\begin{array}{cc} \text{Cov}(Y, Y) & \\ \begin{array}{cc} y & y^2 \\ -2 & 4 \\ 1 & 1 \\ -1 & 1 \\ 2 & 4 \end{array} & \frac{10}{4-1} = \boxed{3.33} \end{array}$$

$$\begin{array}{ccc} \text{Cov}(X, Y) & & \\ \begin{array}{ccc} x & y & x * y \\ -2 & -2 & 4 \\ -1 & 1 & -1 \\ 1 & -1 & -1 \\ 2 & 2 & 4 \end{array} & \frac{6}{4-1} = \boxed{2.0} & \end{array}$$

covariance matrix  $c$

$$\begin{array}{l} \text{var} \rightarrow \\ \text{cov.} \rightarrow \end{array} \left[ \begin{array}{cc} \text{cov}(x, x) = 3.33 & \text{cov}(x, y) = 2.0 \\ \text{cov}(x, y) = 2.0 & \text{cov}(y, y) = 3.33 \end{array} \right] \begin{array}{l} \leftarrow \text{cov.} \\ \leftarrow \text{var} \end{array}$$

$$\mu = 0$$

$$\text{mean}(x_c) = (-2 + (-1) + 1 + 2) / 4 = \frac{0}{4} = 0$$

$$\text{mean}(y_c) = -2 + 1 + (-1) + 2 / 4 = \frac{0}{4} = 0$$

③ Eigenvalues Assume  $\lambda = 1$

$$C.v = \lambda.v$$

$$C - \lambda.I = \begin{vmatrix} 3.33 - \lambda & 2.0 \\ 2.0 & 3.33 - \lambda \end{vmatrix}$$

$$\text{Determinant} = (3.33 - \lambda) \times (3.33 - \lambda) - (2.0) \times (2.0)$$

$$0 = (3.33 - \lambda)^2 - 4$$

$$\Rightarrow (3.33 - \lambda)^2 = 4$$

square root

$$(3.33 - \lambda) = \pm 2.0$$

Two solution -

|                 | calculation  | Eigenvalue  | Role                   |
|-----------------|--------------|-------------|------------------------|
| $\lambda(PC_1)$ | $3.33 + 2.0$ | 5.33        | max var. direction     |
| $\lambda(PC_2)$ | $3.33 - 2.0$ | 1.33        | max var. direction     |
|                 |              | <u>6.66</u> | Total variance in data |

verification: sum of eigenvalue = Trace (C)

$$5.33 + 1.33 = 3.33 + 3.33$$

$$6.66 = 6.66$$

④ eigenvectors

$$(C - \lambda * I) \cdot V = 0$$

$$PC_1 \quad (\lambda_1 = 5.33)$$

$$(C - 5.33 * I) * V = 0$$

$$\begin{vmatrix} 3.33 - 5.33 & 2.0 \\ 2.0 & 3.33 - 5.33 \end{vmatrix} \cdot \begin{vmatrix} V_1 \\ V_2 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \end{vmatrix}$$

$$\text{Row 1: } = -2 * V_1 + 2 * V_2 \Rightarrow V_1 = V_2$$

$$\text{So } V = \begin{vmatrix} 1 \\ 1 \end{vmatrix}$$

Normalize: magnitude =  $\sqrt{1^2 + 1^2} = \sqrt{2}$

$$PC_1 \quad v_1 = \left[ \frac{1}{1.414}, \frac{1}{1.414} \right] = [0.707, 0.707]$$

$$PC_2 \quad (C - 1.33 \cdot I) \cdot v = 0$$

$$\begin{vmatrix} 3.33 - 1.33 & 2.0 \\ 2.0 & 3.33 - 1.33 \end{vmatrix} \cdot \begin{matrix} |v_1| \\ |v_2| \end{matrix} = \begin{matrix} 10 \\ 10 \end{matrix}$$

$$\begin{vmatrix} 2.0 & 2.0 \\ 2.0 & 2.0 \end{vmatrix} \cdot \begin{matrix} |v_1| \\ |v_2| \end{matrix} = \begin{matrix} 10 \\ 10 \end{matrix}$$

$$\text{Row 1: } 2 \cdot v_1 + 2 \cdot v_2 = 0 \quad v_1 = -v_2$$

$$\text{so } v = [-1, 1] \text{ or } [1, -1]$$

$$\text{normalize - magnitude} = \sqrt{(-1)^2 + 1^2} = 1.414$$

$$v_2 = \left[ \frac{-1}{1.414}, \frac{1}{1.414} \right] = [-0.707, 0.707]$$

| PC              | eigenvalue | eigenvector       |                                             |
|-----------------|------------|-------------------|---------------------------------------------|
| PC <sub>1</sub> | 5.33       | $[0.707, 0.707]$  | 45 degree diagonal (x+y)                    |
| PC <sub>2</sub> | 1.33       | $[-0.707, 0.707]$ | 135 degree perpendicular to PC <sub>1</sub> |

$PC_1$  and  $PC_2$  perpendicular (orthogonal) <sup>6</sup>

$$\begin{aligned}\text{Dot product} &= (0.707) * (-0.707) + (0.707) * (0.707) \\ &= -0.5 + 0.5 = 0 \checkmark [\text{correct}]\end{aligned}$$

⑤ Explained variance ratio

$$EVR(PC_k) = \lambda_k / \text{sum}(\text{all } \lambda)$$

$$\text{Total variance} = \lambda_1 + \lambda_2 = 5.33 + 1.33 = 6.66$$

$$EVR(PC_1) = 5.33 / 6.66 = 0.80 = 80\%$$

$$EVR(PC_2) = 1.33 / 6.66 = 0.20 = 20\%$$

$$\text{Cumulative : } PC_1 \text{ alone} = 80\%.$$

$$PC_1 + PC_2 = 100\%.$$

# Dataset - 100

## House price prediction

$x_1$  no. of room

price

$x_2$  Area

$x_3$  carpet Area

$x_4$  super builtup area

wash room

height floor

no. of floor

neigh bar

color of wall

nearest market

nearest school

$x_n$  medical facilities

60%

65%

70%

80%

80%

76%

60%

$f_{10}$

$f_{20}$

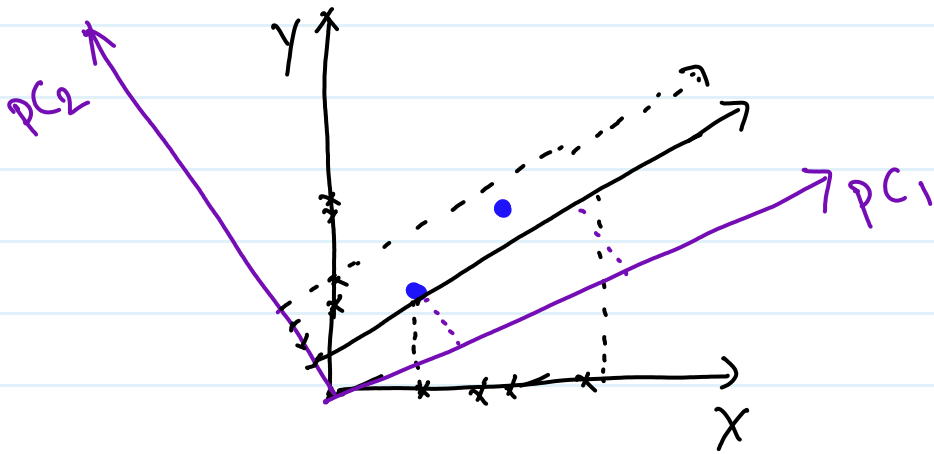
$f_{30}$

$f_{40}$

$f_{50}$

$f_{80}$

$f_{100}$



variance  
eigenvalue  $\rightarrow$  angle  
eigenvector  $\rightarrow$



# # Imbalanced Data

$X_1$   $X_2$   $Y$

Binary class.

- - 1

- - 0

- - 1

class = 1 90%

- - 0

class = 0 10%

- - 1

- - 1

- - 1

train = 1, 1, 1, 1, 1, 1, 1, 0

- - 1

test = 1, 0

- - 1

- - 1

- - ? 1

SMOTE

(Synthetic minority oversampling Technique)

class = 1 8time

class = 0 2time

Applying smote (oversampling)

class = 1 8time

class = 0 8time

# under sampling

1 gtime  
0 2time

1 gtime  
0 2time

# Threshold 0.5  $\begin{matrix} \nearrow 1 \\ \searrow 0 \end{matrix}$

Reduce threshold value

0.25

XGBoost handle imbalanced data